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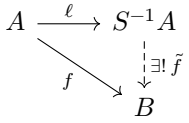
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The Universal Property of Localization

Let A be a commutative ring and $f \in A$. The localization $A_f = A[T]/(fT-1)$ consists of fractions of the form a/f^n for $a \in A$ and $n \geq 0$, with the usual addition and multiplication rules for fractions.

Universal Property: Let $\varphi : A \to B$ be a ring homomorphism such that $\varphi(f)$ is a unit in B. Then there exists a unique ring homomorphism $\tilde{\varphi} : A_f \to B$ making the following diagram commute:



Explicitly, $\tilde{\varphi}$ is given by $\tilde{\varphi}(a/f^n) = \varphi(a) \cdot \varphi(f)^{-n}$.

Proof sketch: The map $a/f^n \mapsto \varphi(a)\varphi(f)^{-n}$ is well-defined because any two representations $a/f^n = a'/f^m$ of the same element satisfy $af^m = a'f^n$ in A, and applying φ gives $\varphi(a)\varphi(f)^m = \varphi(a')\varphi(f)^n$. Since $\varphi(f)$ is invertible in B, we can multiply by $\varphi(f)^{-(n+m)}$ to obtain $\varphi(a)\varphi(f)^{-n} = \varphi(a')\varphi(f)^{-m}$. The homomorphism properties follow directly from the definition, and uniqueness holds because any such homomorphism must satisfy $\tilde{\varphi}(1/f) = \tilde{\varphi}(f/1)^{-1} = \varphi(f)^{-1}$, which determines $\tilde{\varphi}$ on all elements.

Flat Modules

Definition: If R is a ring, then a right R-module A is flat if $A \otimes_R \square$ is an exact functor; that is, whenever

$$0 \to B' \xrightarrow{i} B \xrightarrow{p} B'' \to 0$$

is an exact sequence of left R-modules, then

$$0 \to A \otimes_R B' \xrightarrow{1_A \otimes i} A \otimes_R B \xrightarrow{1_A \otimes p} A \otimes_R B'' \to 0$$

is an exact sequence of abelian groups. Flatness of left R-modules is defined similarly.

Because the functors $A \otimes_R \square : R\text{-Mod} \to \text{Ab}$ are right exact, we see that a right R-module A is flat if and only if, whenever $i : B' \to B$ is an injection, then $1_A \otimes i : A \otimes_R B' \to A \otimes_R B$ is also an injection.

Flat Homomorphisms

A ring homomorphism $\varphi : A \to B$ is called **flat** if B is flat as an A-module (where the A-module structure on B is given by $a \cdot b = \varphi(a)b$ for $a \in A$, $b \in B$).

Equivalently, $\varphi : A \to B$ is flat if the functor $B \otimes_A \square : A\text{-Mod} \to \text{Ab}$ is exact. This means that for every injection $M' \hookrightarrow M$ of A-modules, the induced map $B \otimes_A M' \to B \otimes_A M$ is also injective.

Example: The localization homomorphism $A \to S^{-1}A$ is flat (Corollary 1.2.11).

Lemma 1.2.10

Let A be a ring, S a multiplicative subset of A, and M an A-module. There exists a canonical isomorphism of $S^{-1}A$-modules:

$$M \otimes_A S^{-1}A \simeq S^{-1}M$$

Proof: We have a canonical homomorphism of A-modules $M \otimes_A S^{-1}A \to S^{-1}M$ defined by $m \otimes (a/s) \mapsto (am)/s$. This is an isomorphism because it has an inverse given by $m/s \mapsto m \otimes (1/s)$ (this map is well defined). Moreover, this homomorphism is clearly compatible with the structure of $S^{-1}A$-modules.

Corollary 1.2.11

For any multiplicative subset S of A, the canonical homomorphism $A \to S^{-1}A$ is flat.

Proof: Let $N \subseteq M$ be an A-module inclusion. By construction, $S^{-1}N$ is a submodule of $S^{-1}M$. Hence the induced map $N \otimes_A S^{-1}A \to M \otimes_A S^{-1}A$ is injective, which shows $S^{-1}A$ is flat over A.

Lemma 2.7

Let $\mathcal{U} = \{U_i\}_i$ be a family of open subsets of X. Let $U = \bigcup_i U_i$ and $U_{ij} = U_i \cap U_j$. For any presheaf $\mathcal{F}$ on X, we have a complex of Abelian groups $C_\bullet(\mathcal{U}, \mathcal{F})$:

$$0 \to \mathcal{F}(U) \xrightarrow{d_0} \prod_i \mathcal{F}(U_i) \xrightarrow{d_1} \prod_{i,j} \mathcal{F}(U_{ij})$$

defined by $d_0 : s \mapsto (s|_{U_i})_i$ and $d_1 : (s_i)_i \mapsto (s_i|_{U_{ij}} - s_j|_{U_{ij}})_{i,j}$.

Lemma 2.7: With the above notation, $\mathcal{F}$ is a sheaf if and only if $C_\bullet(\mathcal{U}, \mathcal{F})$ is exact for every family of open subsets $\mathcal{U}$ of X.

Proof: We note that the injectivity of d_0 is equivalent to condition (4) of Definition 2.2 (locality/uniqueness), while the equality $\ker d_1 = \operatorname{im} d_0$ is equivalent to condition (5) (gluing).

Exercise 2.7

Let $\mathcal{B}$ be a base of open subsets on a topological space X. Let $\mathcal{F}, \mathcal{G}$ be two sheaves on X. Suppose that for every $U \in \mathcal{B}$ there exists a homomorphism $\alpha(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ which is compatible with restrictions. Show that this extends in a unique way to a homomorphism of sheaves $\alpha : \mathcal{F} \to \mathcal{G}$. Show that if $\alpha(U)$ is surjective (resp. injective) for every $U \in \mathcal{B}$, then α is surjective (resp. injective).

Solution:

Extension to a sheaf homomorphism

For any open set $U \subseteq X$, write U as a union of basic open sets from $\mathcal{B}$: $U = \bigcup_{i \in I} U_i$ with $U_i \in \mathcal{B}$. By the sheaf property of $\mathcal{F}$ and $\mathcal{G}$, we have:

$$\mathcal{F}(U) = \ker\left(\prod_{i \in I} \mathcal{F}(U_i) \to \prod_{i,j \in I} \mathcal{F}(U_i \cap U_j)\right)$$

Define $\alpha(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ as follows: for $s \in \mathcal{F}(U)$, let $\alpha(U)(s)$ be the unique section in $\mathcal{G}(U)$ whose restriction to each U_i equals $\alpha(U_i)(s|_{U_i})$. This is well-defined because the maps $\alpha(U_i)$ are compatible with restrictions:

$$\alpha(U_i)(s|_{U_i})|_{U_i \cap U_j} = \alpha(U_i \cap U_j)(s|_{U_i \cap U_j}) = \alpha(U_j)(s|_{U_j})|_{U_i \cap U_j}$$

so the sections $\alpha(U_i)(s|_{U_i})$ glue to a unique section of $\mathcal{G}(U)$.

Uniqueness: Any sheaf homomorphism extending the $\alpha(U)$ must satisfy the gluing condition above, so the construction is forced.

Surjectivity

If each $\alpha(U)$ for $U \in \mathcal{B}$ is surjective, let $t \in \mathcal{G}(V)$ for any open $V \subseteq X$. Write $V = \bigcup_i V_i$ with $V_i \in \mathcal{B}$. For each i, surjectivity of $\alpha(V_i)$ gives $s_i \in \mathcal{F}(V_i)$ with $\alpha(V_i)(s_i) = t|_{V_i}$.

On overlaps $V_i \cap V_j$, we have $\alpha(V_i \cap V_j)(s_i|_{V_i \cap V_j} - s_j|_{V_i \cap V_j}) = 0$. Since $\alpha(V_i \cap V_j)$ is injective on $V_i \cap V_j \in \mathcal{B}$ (assuming $\mathcal{B}$ is closed under finite intersections, or working with a basis), we get $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$. By the sheaf property of $\mathcal{F}$, the s_i glue to $s \in \mathcal{F}(V)$ with $\alpha(V)(s) = t$.

Injectivity

If each $\alpha(U)$ for $U \in \mathcal{B}$ is injective and $s \in \mathcal{F}(V)$ satisfies $\alpha(V)(s) = 0$, then for any $U \in \mathcal{B}$ with $U \subseteq V$, we have $\alpha(U)(s|_U) = 0$. By injectivity of $\alpha(U)$, $s|_U = 0$. Since $\mathcal{B}$ is a basis, this implies $s = 0$ in $\mathcal{F}(V)$ (a section that vanishes on a basis must be zero by the sheaf property).

Lemma 2.9

Let $\mathcal{F}$ be a sheaf on X. Let $s, t \in \mathcal{F}(X)$ be sections such that $s_x = t_x$ for every $x \in X$. Then $s = t$.

Proof. We may assume that $t = 0$. For every $x \in X$, there exists an open neighborhood U_x of x such that $s|_{U_x} = 0$, because $s_x = 0$. As the U_x cover X when x varies, we indeed have $s = 0$.

Proposition 2.12

Let $\alpha : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves on X. Then α is an isomorphism if and only if α_x is an isomorphism for every $x \in X$.

Proof. Suppose that α_x is an isomorphism for every $x \in X$. Let us show that α is an isomorphism (the converse is trivial). Let $s \in \mathcal{F}(U)$ be a section. If $\alpha(U)(s) = 0$, then for every $x \in U$, we have $\alpha_x(s_x) = (\alpha(U)(s))_x = 0$. It follows that $s_x = 0$ for every $x \in U$. Hence $s = 0$ (Lemma 2.9) and α is injective. Next, let $t \in \mathcal{G}(U)$. Then we can find a covering of U by open sets U_i and sections $s_i \in \mathcal{F}(U_i)$ such that $\alpha(U_i)(s_i) = t|_{U_i}$. As we have just seen that α is injective, s_i and s_j coincide on $U_i \cap U_j$. The s_i therefore glue to a section $s \in \mathcal{F}(U)$ (i.e., $s|_{U_i} = s_i$). By construction, $\alpha(U)(s)$ and t coincide on every U_i, and are therefore equal. This proves that $\alpha(U)$ is surjective.

Topological Open Immersion

Definition 2.22. Let $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. We say that:

- $(f, f^\#)$ is an **open immersion** if:
  1. $f : X \to Y$ is a topological open immersion (i.e., f is a homeomorphism of X onto an open subset $V \subseteq Y$)
  2. For every $x \in X$, the stalk map $f_x^\# : \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ is an isomorphism
- $(f, f^\#)$ is a **closed immersion** if:
  1. $f : X \to Y$ is a topological closed immersion (i.e., f is a homeomorphism of X onto a closed subset $V \subseteq Y$)
  2. For every $x \in X$, the stalk map $f_x^\# : \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ is surjective

Characterization of Open Immersions

Let V be an open subset of Y. Then $(V, \mathcal{O}_Y|_V)$ is a ringed topological space, where the structure sheaf $\mathcal{O}_Y|_V$ is defined by $(\mathcal{O}_Y|_V)(U) = \mathcal{O}_Y(U)$ for every open $U \subseteq V$. In all that follows, an open subset V will always be endowed with this structure induced by $\mathcal{O}_Y$.

Key fact: $(f, f^\#)$ is an open immersion if and only if there exists an open subset $V \subseteq Y$ such that $(f, f^\#)$ induces an isomorphism of ringed spaces:

$$(X, \mathcal{O}_X) \simeq (V, \mathcal{O}_Y|_V)$$

This characterization follows from Proposition 2.12: a morphism is an isomorphism precisely when it induces isomorphisms on all stalks.

In Plain English

What is an open immersion?

Think of an open immersion as identifying one scheme (or ringed space) as an "open piece" of another. Just as an open interval $(0, 1)$ sits naturally inside the real line $\mathbb{R}$, an open immersion describes when one scheme X can be viewed as an open subscheme of another scheme Y.

The two conditions:

1. **Topological condition:** The map $f : X \to Y$ must be a homeomorphism onto its image, and that image must be open in Y. This means X looks exactly like some open region V of Y from a purely topological perspective—same "shape," same open sets.
2. **Algebraic condition:** At the level of functions (the stalk maps), the structure must be preserved perfectly. Every function near $f(x)$ in Y should correspond uniquely to a function near x in X, and vice versa.

Why we need both conditions:

The topological condition ensures X "sits inside" Y as an open region in the right way. The algebraic condition ensures the functions match up correctly—we're not just embedding the space, but also its structure sheaf of regular functions.

Example:

Consider $X = \mathrm{Spec}(k[x])$ (the affine line) and $Y = \mathrm{Spec}(k[x, y])$ (the affine plane). The map $X \to Y$ induced by $k[x, y] \to k[x]$ sending $x \mapsto x$, $y \mapsto 0$ is a closed immersion—X becomes a line inside the plane.

For an open immersion, consider the inclusion $D(x) \hookrightarrow \mathrm{Spec}(k[x])$. This identifies the "punctured line" (the line with the origin removed) as an open subscheme of the full line.

Connection to Proposition 3.12:

In Lemma 3.7, we use the fact that the localization morphism $A \to A_g$ induces an open immersion

$$\mathrm{Spec}(A_g) \hookrightarrow \mathrm{Spec}(A)$$

This is exactly what allows us to identify $D(f)$ (where f doesn't vanish) with an affine scheme $\mathrm{Spec}(A_f)$.

Lemma 1.7

Let $\phi : A \to B$ be a ring homomorphism. Then we have a map of sets $\mathrm{Spec}(\phi) : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ defined by $\mathfrak{p} \mapsto \phi^{-1}(\mathfrak{p})$ for every $\mathfrak{p} \in \mathrm{Spec}(B)$. Let $f = \mathrm{Spec}(\phi)$ be the map associated to ϕ as above. The following properties are true:

(a) The map f is continuous.
(b) If ϕ is surjective, then f induces a homeomorphism from $\mathrm{Spec}(B)$ onto the closed subset $V(\mathrm{Ker}(\phi))$ of $\mathrm{Spec}(A)$.
(c) If ϕ is a localization morphism $A \to S^{-1}A$, then f is a homeomorphism from $\mathrm{Spec}(S^{-1}A)$ onto the subspace $\{\mathfrak{p} \in \mathrm{Spec}(A) \mid \mathfrak{p} \cap S = \emptyset\}$ of $\mathrm{Spec}(A)$.

Proof:

(a) Let I be an ideal of A. We let IB denote the ideal of B generated by $\phi(I)$. It is easy to see that $f^{-1}(V(I)) = V(IB)$. Hence f is continuous.

(b) If ϕ is surjective, f clearly establishes a continuous bijection from $\mathrm{Spec}(B)$ onto $V(\mathrm{Ker}(\phi))$. Moreover, it sends a closed set $V(J)$ onto a closed set $V(\phi^{-1}(J))$. It follows that f is closed, and therefore a homeomorphism.

(c) The assertion concerning the image of f, as well as the injectivity of f, follows directly from the definition (see 1.2.2). Let $B = S^{-1}A$, and let J be an ideal of B. Then we easily see that $f(V(J)) = V(\phi^{-1}(J)) \cap f(\mathrm{Spec}(B))$, which shows that f is a closed map.

Lemma 3.7

Let $X = Spec(A)$ be an affine scheme, and let $g \in A$. Then the open subset $D(g)$, endowed with the structure of a ringed topological space induced by that of X, is an affine scheme isomorphic to $Spec(A_g)$ (as a ringed topological space).

Understanding the Statement

What is $D(g)$?

For a ring element $g \in A$, the distinguished open set $D(g) \subseteq \mathrm{Spec}(A)$ is defined as:

$$D(g) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid g \notin \mathfrak{p}\}$$

Geometrically, this is the set of prime ideals that **don't** contain g — in other words, the "locus where g doesn't vanish."

What is $\mathcal{O}_X|_{D(g)}$?

The notation $\mathcal{O}_X|_{D(g)}$ denotes the **restriction of the structure sheaf** $\mathcal{O}_X$ to the open subset $D(g)$. Formally, this is a sheaf on $D(g)$ defined by:

$$(\mathcal{O}_X|_{D(g)})(U) = \mathcal{O}_X(U) \quad \text{for every open } U \subseteq D(g)$$

The pair $(D(g), \mathcal{O}_X|_{D(g)})$ is a ringed topological space — it's the scheme X "viewed only on the open set where g doesn't vanish."

Proof

Let Y denote the affine scheme $\mathrm{Spec}(A_g)$. By Lemma 1.7(c), the localization homomorphism $A \to A_g$ induces a topological open immersion

$$i : Y = \mathrm{Spec}(A_g) \hookrightarrow \mathrm{Spec}(A) = X$$

whose image is exactly $D(g)$.

Why is $\mathrm{Spec}(A_g)$ homeomorphic to $D(g)$?

By Lemma 1.7(c), the map $\mathrm{Spec}(A_g) \to \mathrm{Spec}(A)$ is a homeomorphism onto the subspace

$$\{\mathfrak{p} \in \mathrm{Spec}(A) \mid \mathfrak{p} \cap S = \emptyset\}$$

where $S = \{1, g, g^2, g^3, ...\}$. But $\mathfrak{p} \cap S = \emptyset$ means precisely that $g^n \notin \mathfrak{p}$ for all $n \geq 0$, which is equivalent to $g \notin \mathfrak{p}$. Hence this subspace is $D(g)$.

Checking the sheaf condition:

To show that i is an isomorphism of ringed spaces $(Y, \mathcal{O}_Y) \simeq (D(g), \mathcal{O}_X|_{D(g)})$, we need to verify that the structure sheaves match up. By Proposition 2.12 and Exercise 2.7, it suffices to check this on a basis of open sets.

Let $D(h)$ be a principal open subset of $X = \mathrm{Spec}(A)$ contained in $D(g)$. Let $\bar{h}$ be the image of h in A_g. Then:

$$\begin{aligned}
\mathcal{O}_X(D(h)) &= A_h \\
&\simeq (A_g)_{\bar{h}} \quad \text{(localization commutes with localization)} \\
&= \mathcal{O}_Y(D(\bar{h})) \\
&= (i_*\mathcal{O}_Y)(D(h))
\end{aligned}$$

Why does $A_h \simeq (A_g)_{\bar{h}}$?

Localizing at g and then at $\bar{h}$ is the same as localizing at gh:

$$(A_g)_{\bar{h}} = A_g[\bar{h}^{-1}] = A[(g^{-1})(h^{-1})] = A[(gh)^{-1}] = A_{gh}$$

Since $D(h) \subseteq D(g)$, any prime ideal not containing h also doesn't contain g, so $D(h) = D(gh)$ and therefore $A_h = A_{gh}$.

Conclusion:

Since the principal open sets $D(h)$ form a basis for the topology on $D(g)$, and we've shown $\mathcal{O}_X|_{D(g)}$ agrees with $i_*\mathcal{O}_Y$ on this basis, Exercise 2.7 tells us the two sheaves are isomorphic. Therefore i induces an isomorphism of ringed spaces:

$$(Y, \mathcal{O}_Y) = \mathrm{Spec}(A_g) \simeq (D(g), \mathcal{O}_X|_{D(g)})$$

In Plain English

Geometric meaning:

Lemma 3.7 tells us that "localizing the ring" and "restricting to where g doesn't vanish" are two sides of the same coin:

- **Algebra side:** $A \to A_g$ inverts the element g
- **Geometry side:** $\mathrm{Spec}(A) \to \mathrm{Spec}(A_g)$ restricts to the open set $D(g)$

The lemma guarantees these operations are compatible: the scheme you get by localizing the ring is exactly the same as the open subscheme where g is invertible.

Example:

Take $A = k[x]$ (the coordinate ring of the affine line) and $g = x$. Then: * $D(x)$ is the affine line with the origin removed * $A_x = k[x]_x = k[x, x^{-1}]$ is the ring of Laurent polynomials

The lemma says:

$$\mathrm{Spec}(k[x, x^{-1}]) \simeq D(x) \subset \mathrm{Spec}(k[x])$$

This makes geometric sense: the ring $k[x, x^{-1}]$ consists of functions that may have negative powers of x (they're allowed to "blow up" at $x = 0$), which is exactly what you need for functions on the line with the origin removed.

Definition 3.11

Let X be a scheme and $f \in \mathcal{O}_X(X)$. We denote by X_f the set of $x \in X$ such that $f_x \in \mathcal{O}_{X,x}^*$ (i.e., f_x is a unit in the stalk at x).

Let us consider the following condition:

$$\begin{cases} X \text{ admits a covering by a finite number of affine} \\ \text{open subsets } \{U_i\}_i \text{ such that } U_i \cap U_j \text{ also admits} \\ \text{a finite covering by affine open subsets.} \end{cases} \tag{3.2}$$

Affine schemes always verify this condition, as do Noetherian schemes (Proposition 3.46(a)).

Proposition 3.12

Let X be a scheme and $f \in \mathcal{O}_X(X)$. Then X_f is an open subset of X. Moreover, if X verifies condition (3.2) above, then the restriction $\mathcal{O}_X(X) \to \mathcal{O}_X(X_f)$ induces an isomorphism $\mathcal{O}_X(X)_f \simeq \mathcal{O}_X(X_f)$.

$\mathcal{O}_X(X) \to \mathcal{O}_X(X_f)$ induces $\alpha : \mathcal{O}_X(X)_f \to \mathcal{O}_X(X_f)$

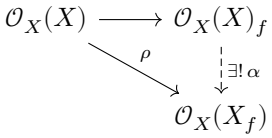
If $x \in X_f$, then by definition f_x is invertible in the stalk $\mathcal{O}_{X,x}$. Thus there exists an open neighborhood $U \ni x$ and a $g \in \mathcal{O}_X(U)$ such that $f|_x g|_x = 1$ in $\mathcal{O}_{X,x}$. Since equality in the stalk means equality on some open neighborhood, it follows that $(fg)|_V = 1$ in an open neighborhood $V \subseteq U$ of x.

Why $V \subseteq X_f$: For any point $y \in V$, the section $(fg)|_V = 1$ restricts to $f_y g_y = 1$ in the stalk $\mathcal{O}_{X,y}$. This shows f_y is invertible (with inverse g_y), hence $y \in X_f$ by definition. Therefore $V \subseteq X_f$.

Why X_f is open: We have shown that every point $x \in X_f$ admits an open neighborhood V such that $V \subseteq X_f$. This means $X_f = \bigcup_{x \in X_f} V_x$ is a union of open sets, hence X_f itself is open. Moreover, by glueing the inverses $g \in \mathcal{O}_X(V)$ of f as the V vary, we obtain an element of $\mathcal{O}_X(X_f)$, the inverse of $f|_{X_f}$.

Consequently, $\mathcal{O}_X(X) \to \mathcal{O}_X(X_f)$ induces a homomorphism $\alpha : \mathcal{O}_X(X)_f \to \mathcal{O}_X(X_f)$.

Why α exists: The restriction map $\rho : \mathcal{O}_X(X) \to \mathcal{O}_X(X_f)$ sends $f \mapsto f|_{X_f}$. Since we constructed an inverse of $f|_{X_f}$ in $\mathcal{O}_X(X_f)$ by glueing the local inverses g, the element $f|_{X_f}$ is a unit in $\mathcal{O}_X(X_f)$. By the universal property of localization, any ring homomorphism $\varphi : A \to B$ with $\varphi(f)$ a unit factors uniquely through A_f. Applying this to $\rho : \mathcal{O}_X(X) \to \mathcal{O}_X(X_f)$, we obtain the unique homomorphism $\alpha : \mathcal{O}_X(X)_f \to \mathcal{O}_X(X_f)$ satisfying $\alpha(a/1) = \rho(a)$ for all $a \in \mathcal{O}_X(X)$. Explicitly, $\alpha(a/f^n) = \rho(a) \cdot (\rho(f))^{-n}$.



α is Iso

By hypothesis (3.2), X admits a finite covering $\mathcal{U} = \{U_i\}_i$ by affine open subsets. Hence X_f is the union of the $V_i := U_i \cap X_f = D(f|_{U_i})$.

Note: in a topological space if X is the union of a finite covering of U_i of open subsets, then if Y is an open subset of X, then Y is the union of $V_i = Y \cap U_i$.

Why $V_i = D(f|_{U_i})$: Since each $U_i = \mathrm{Spec}(A_i)$ is affine, the distinguished open $D(f|_{U_i}) \subseteq \mathrm{Spec}(A_i)$ consists precisely of the primes $\mathfrak{p} \subset A_i$ where $f \notin \mathfrak{p}$, which is equivalent to f being a unit in the stalk $A_{i,\mathfrak{p}}$. This is exactly the definition of $U_i \cap X_f$.

$$\implies V_i := U_i \cap X_f = D(f|_{U_i})$$

(since V_i is affine)

For simplicity, let us also denote by f its restriction to any open subset of X.

We have $\mathcal{O}_X(U_i)_f = \mathcal{O}_X(V_i)$ by virtue of Lemma 3.7.

Detailed explanation:

Recall $V_i = U_i \cap X_f = D(f|_{U_i})$. Since $U_i = \mathrm{Spec}(A_i)$ is affine with $A_i = \mathcal{O}_X(U_i)$, Lemma 3.7 tells us:

$$D(f|_{U_i}) \cong \mathrm{Spec}(\mathcal{O}_X(U_i)_f)$$

as ringed topological spaces. By the very definition of the structure sheaf on an affine scheme, the ring of regular functions on $\mathrm{Spec}(\mathcal{O}_X(U_i)_f)$ is precisely $\mathcal{O}_X(U_i)_f$. Therefore:

$$\mathcal{O}_X(V_i) = \mathcal{O}_X(D(f|_{U_i})) = \mathcal{O}_X(U_i)_f$$

With the notation of Lemma 2.7, we have an exact complex $C_\bullet(\mathcal{U}, \mathcal{O}_X)$ where we can replace $\prod$ by $\oplus$ since there are only a finite number of terms:

$$0 \to \mathcal{O}_X(X) \xrightarrow{d_0} \bigoplus_i \mathcal{O}_X(U_i) \xrightarrow{d_1} \bigoplus_{i,j} \mathcal{O}_X(U_i \cap U_j)$$

where d_0 sends $s \mapsto (s|_{U_i})_i$ and d_1 sends $(s_i)_i \mapsto (s_i|_{U_i \cap U_j} - s_j|_{U_i \cap U_j})_{i,j}$.

Therefore $C_\bullet(\mathcal{U}, \mathcal{O}_X) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(X)_f$ is still exact owing to the flatness of $\mathcal{O}_X(X) \to \mathcal{O}_X(X)_f$ (Corollary 1.2.11).

Here the functor is $\square \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(X)_f$. For any $\mathcal{O}_X(X)$-module M since we have the localisation homomorphism $\mathcal{O}_X(X) \to \mathcal{O}_X(X)_f$, so $M \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(X)_f \cong M_f$ by $M \otimes_A S^{-1}A \cong S^{-1}A$.

$$\begin{aligned}
\mathcal{O}_X(X) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(X)_f &\cong \mathcal{O}_X(X)_f, \\
\bigoplus_i \mathcal{O}_X(U_i) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(X)_f &\cong \bigoplus_i \left(\mathcal{O}_X(U_i) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(X)_f \right), \\
&\cong \bigoplus_i \mathcal{O}_X(U_i)_f, \\
\bigoplus_{i,j} \mathcal{O}_X(U_i \cap U_j) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(X)_f &\cong \bigoplus_{i,j} \mathcal{O}_X(U_i \cap U_j)_f.
\end{aligned}$$

Explanation of the second line: The tensor product distributes over direct sums, giving $\bigoplus_i \left(\mathcal{O}_X(U_i) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(X)_f \right)$. Each $\mathcal{O}_X(U_i)$ is an $\mathcal{O}_X(X)$-algebra via the restriction map, and localization commutes with tensor products: $\mathcal{O}_X(U_i) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(X)_f \cong \mathcal{O}_X(U_i)_f$ (where the right side means localization at the image of f in $\mathcal{O}_X(U_i)$).

This gives a commutative diagram:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{O}_X(X)_f & \xrightarrow[d_0]{} & \bigoplus_i \mathcal{O}_X(U_i)_f & \xrightarrow[d_1]{} & \bigoplus_{i,j} \mathcal{O}_X(U_i \cap U_j)_f & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \simeq} & & \downarrow{\scriptstyle \beta} & & \\
0 & \longrightarrow & \mathcal{O}_X(X_f) & \xrightarrow[d_0]{} & \bigoplus_i \mathcal{O}_X(V_i) & \xrightarrow[d_1]{} & \bigoplus_{i,j} \mathcal{O}_X(V_i \cap V_j) & \longrightarrow & 0
\end{array}$$

where the horizontal rows are exact.

Why α is injective: Let $a \in \mathcal{O}_X(X)_f$ with $\alpha(a) = 0$. By exactness of the top row, a maps to some element $b \in \bigoplus_i \mathcal{O}_X(U_i)_f$. By commutativity of the left square, b maps to 0 in $\bigoplus_i \mathcal{O}_X(V_i)$.

We showed above $\bigoplus_i \mathcal{O}_X(U_i)_f = \bigoplus_i \mathcal{O}_X(V_i)$ (see "detailed explanation"). Since b maps to 0 under this isomorphism, we must have $b = 0$.

By exactness of the top row, the first map $\mathcal{O}_X(X)_f \to \bigoplus_i \mathcal{O}_X(U_i)_f$ is injective, so $a = 0$. Hence α is injective.

Why β is injective:

By hypothesis (3.2), each intersection $U_i \cap U_j$ admits a finite covering by affine open subsets. Let $\{W_{ijk}\}_k$ be such a finite affine open covering of $U_i \cap U_j$. Then we can apply the same Čech complex construction to each $U_i \cap U_j$:

$$0 \to \mathcal{O}_X(U_i \cap U_j) \to \bigoplus_k \mathcal{O}_X(W_{ijk}) \to \bigoplus_{k,\ell} \mathcal{O}_X(W_{ijk} \cap W_{ij\ell}) \to 0$$

Localizing at f (using flatness), we get:

$$0 \to \mathcal{O}_X(U_i \cap U_j)_f \to \bigoplus_k \mathcal{O}_X(W_{ijk})_f \to \bigoplus_{k,\ell} \mathcal{O}_X(W_{ijk} \cap W_{ij\ell})_f \to 0$$

But $W_{ijk} \subseteq U_i \cap U_j \subseteq X_f$, so $W_{ijk} \cap X_f = W_{ijk}$, and by Lemma 3.7, $\mathcal{O}_X(W_{ijk})_f \cong \mathcal{O}_X(W_{ijk})$. Similarly, $(W_{ijk} \cap W_{ij\ell})_f = W_{ijk} \cap W_{ij\ell}$ and $\mathcal{O}_X(W_{ijk} \cap W_{ij\ell})_f \cong \mathcal{O}_X(W_{ijk} \cap W_{ij\ell})$.

This gives us a commutative diagram for each pair (i, j):

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{O}_X(U_i \cap U_j)_f & \longrightarrow & \bigoplus_k \mathcal{O}_X(W_{ijk})_f & \longrightarrow & \bigoplus_{k,\ell} \mathcal{O}_X(W_{ijk} \cap W_{ij\ell})_f & \longrightarrow & 0 \\ & & \downarrow & & \simeq\downarrow & & \simeq\downarrow & & \\ 0 & \longrightarrow & \mathcal{O}_X(V_i \cap V_j) & \longrightarrow & \bigoplus_k \mathcal{O}_X(W_{ijk}) & \longrightarrow & \bigoplus_{k,\ell} \mathcal{O}_X(W_{ijk} \cap W_{ij\ell}) & \longrightarrow & 0 \end{array}$$

By the exact same reasoning as for α, the leftmost vertical map (which is precisely the component of β corresponding to (i, j)) is injective. Since this holds for all pairs (i, j), the direct sum map β is injective.

Why α is an isomorphism given that α and β are injective:

We need to show α is surjective. Let $y \in \mathcal{O}_X(X_f)$. By exactness of the bottom row, there exists an element $(s_i) \in \bigoplus_i \mathcal{O}_X(V_i)$ mapping to y. Since the middle vertical map is surjective onto $\bigoplus_i \mathcal{O}_X(V_i)$ (as each $\mathcal{O}_X(U_i)_f \to \mathcal{O}_X(V_i)$ is an isomorphism by Lemma 3.7), we can lift (s_i) to some $(t_i) \in \bigoplus_i \mathcal{O}_X(U_i)_f$.

By commutativity of the right square, (t_i) maps to 0 in $\bigoplus_{i,j} \mathcal{O}_X(V_i \cap V_j)$ (since y maps to 0 by exactness). Since β is injective, (t_i) must map to 0 in $\bigoplus_{i,j} \mathcal{O}_X(U_i \cap U_j)_f$. By exactness of the top row, there exists $x \in \mathcal{O}_X(X)_f$ mapping to (t_i). By commutativity of the left square, $\alpha(x) = y$. Hence α is surjective.

Since α is both injective and surjective, it is an isomorphism.

Example: The Projective Line $\mathbb{P}^1_k$

Let $X = \mathbb{P}^1_k$ be the projective line over a field k, with homogeneous coordinates $[x_0 : x_1]$.

Standard affine open covering

$$\mathcal{U} = \{U_0, U_1\}, \quad \text{where } U_0 = \{x_0 \neq 0\} \cong \mathbb{A}^1_k, \quad U_1 = \{x_1 \neq 0\} \cong \mathbb{A}^1_k$$

We have:

- $\mathcal{O}_X(U_0) = k[t]$ where $t = x_1/x_0$
- $\mathcal{O}_X(U_1) = k[s]$ where $s = x_0/x_1$
- $\mathcal{O}_X(U_0 \cap U_1) = k[t, t^{-1}] \cong k[s, s^{-1}]$ with $s = t^{-1}$

Choosing a global section

Let $f = x_0 \in H^0(\mathbb{P}^1, \mathcal{O}(1))$. However, this is a section of $\mathcal{O}(1)$, not $\mathcal{O}_X$. Instead, let's work with $f \in \mathcal{O}_X(X) = k$ (the only global functions are constants).

Take $f = 2 \in k$ (assuming $\text{char}(k) \neq 2$). Then:

- $X_f = \{p \in \mathbb{P}^1 \mid f_p \text{ is a unit in } \mathcal{O}_{X,p}\} = \mathbb{P}^1$ (since 2 is invertible in k)

The proposition gives:

$$\mathcal{O}_X(X)_f = k_2 \cong k \simeq \mathcal{O}_X(X_f) = k$$

A more interesting case: $X = \mathbf{Spec}(A) \cup \mathbf{Spec}(B)$

Let $A = k[x]$, $B = k[y]$, and glue along $D(x) \cong D(y)$ via $x \leftrightarrow y^{-1}$. This gives $\mathbb{P}^1_k$ as before.

Now consider f defined as x on $U_0 = \mathrm{Spec}(A)$ and 1 on $U_1 = \mathrm{Spec}(B)$.

Check compatibility on overlap: On $U_0 \cap U_1$, we have $x = y^{-1}$, and $1 = 1$. These don't match, so this isn't a valid global section.

Example with a genuine non-constant function on a non-separated scheme

Let X be the affine line with doubled origin. Let $A = k[t]$, and let $X = \mathrm{Spec}(A) \cup_{\mathrm{Spec}(A[t^{-1}])} \mathrm{Spec}(A)$.

Construction: We take two copies of $\mathbb{A}^1 = \mathrm{Spec}(k[t])$ and glue them along the open subset $\mathrm{Spec}(k[t, t^{-1}]) = \mathbb{A}^1 \setminus \{0\}$. This means the origins are **not** identified — we have two distinct points 0_1 and 0_2.

Take $f = t \in A$. Then:

- $X_f = D(t)$ in both copies of $\mathrm{Spec}(A)$, which are identified
- $X_f = \mathbb{A}^1_k \setminus \{0\}$

Why the origin is removed from both copies:

The stalk at the origin in either copy is $\mathcal{O}_{X,0_i} = k[t]_{(t)}$ — the ring of rational functions where the denominator doesn't vanish at $t = 0$. In this stalk, t is **not** a unit because:

$$t \cdot \left(\frac{a}{b}\right) = 1 \implies ta = b$$

But if $a, b \in k[t]$ with $b(0) \neq 0$, then $ta(0) = 0 \neq b(0)$. So t cannot be inverted in the stalk.

Therefore 0_1 and 0_2 are both **not** in X_f, since $f = t$ fails to be a unit in both stalks. Geometrically, X_f consists of both copies of $\mathbb{A}^1$ with their origins removed, glued along the overlap — which gives us back $\mathbb{A}^1 \setminus \{0\}$.

The proposition gives:

$$\mathcal{O}_X(X)_f = k[t]_t \cong k[t, t^{-1}] \simeq \mathcal{O}_X(X_f) = k[t, t^{-1}]$$

Verification

On $X_f = \mathbb{A}^1_k \setminus \{0\} = \mathrm{Spec}(k[t, t^{-1}])$:

- $\mathcal{O}_X(X_f) = k[t, t^{-1}]$
- $\mathcal{O}_X(X)_f = k[t]_t = k[t, t^{-1}]$

The isomorphism is indeed $\frac{p(t)}{t^n} \mapsto \frac{p(t)}{t^n}$.

Why This Proposition Matters

The Big Picture

Proposition 3.12 establishes a fundamental bridge between **algebra** and **geometry** in scheme theory. It tells us that:

1. **Geometric side**: The set X_f — where a function f doesn't vanish (is invertible in the stalks) — forms an open subset of the scheme
2. **Algebraic side**: The ring of functions on this open set is just the localization of the global functions at f

In plain English: If you want to study your scheme where $f \neq 0$, you can algebraically localize the ring of global functions at f. This is incredibly powerful because it lets us translate geometric problems into algebraic computations.

Why X_f instead of $D(f)$?

For an **affine scheme** $X = \mathrm{Spec}(A)$ and $f \in A$, the two notions coincide:

$$X_f = D(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid f \notin \mathfrak{p}\}$$

However, for **general schemes**, they differ:

Notation	When defined	Meaning
$D(f)$	Only for affine $\mathrm{Spec}(A)$	Primes not containing f
X_f	Any scheme, any $f \in \mathcal{O}_X(X)$	Points where f is a unit in the stalk

Key distinction: X_f is defined for *any* scheme and *any* global section, not just affine ones. For non-affine schemes (like $\mathbb{P}^1$), there's no single ring A where we can talk about $D(f)$ in the usual sense.

When are they different?

Example 1: Projective space $\mathbb{P}^1$

The global functions on $\mathbb{P}^1$ are just constants: $\mathcal{O}_X(X) = k$. If $f = 2 \in k$, then: - $X_f = \mathbb{P}^1$ (since 2 is invertible everywhere) - $\mathcal{O}_X(X)_f = k_2 \cong k = \mathcal{O}_X(X_f)$

This would be meaningless if we tried to write $D(2)$ — there's no single $\mathrm{Spec}(A)$ here!

Example 2: Non-separated schemes (affine line with doubled origin)

Here $X = \mathrm{Spec}(k[t]) \cup_{\mathrm{Spec}(k[t,t^{-1}])} \mathrm{Spec}(k[t])$. The function $f = t$ is a global section (it's t on both copies). The set X_f is $\mathbb{A}^1 \setminus \{0\}$ — the origin is removed from **both** copies. This natural geometric notion is captured by X_f, not by any $D(f)$.

Practical Applications

1. Computing with sheaves

When you need to compute cohomology or other invariants of a scheme, you can work with localized rings instead of directly with open sets. For example, on $\mathbb{P}^n$ covered by affine opens $D_+(x_i)$, the transition functions are localizations.

2. Gluing local data

The proposition assures us that localizing at f and then restricting to an affine open U_i gives the same result as restricting first and then localizing. This compatibility is essential for patching together local constructions.

3. Studying degenerations

If f vanishes along some subvariety (a "divisor"), then X_f is the complement. The proposition tells us the algebraic structure of this complement is just $\mathcal{O}_X(X)_f$. This is used constantly in:

- **Resolution of singularities**: Study what happens when you remove the singular locus
- **Moduli problems**: Construct parameter spaces by removing bad loci
- **D-module theory**: Study differential equations away from singularities

4. Computational algebraic geometry

In computer algebra systems like Macaulay2 or Sage, when you work with quotients and localizations, this proposition guarantees that the geometric interpretation matches the algebraic computation.

The Takeaway

Proposition 3.12 is the **scheme-theoretic generalization** of the basic fact that:

$$\text{The ring of regular functions on } D(f) \subseteq \mathrm{Spec}(A) \text{ is } A_f$$

It tells us this works for *any* scheme (not just affine ones), provided we have a finite affine open covering. This is a foundational tool that makes scheme theory computationally tractable — we can often reduce global problems to local algebra on affine patches, then glue the results back together.

Computing in Sage

Here's how to verify Proposition 3.12 for the projective line example in SageMath:

```
# Define the base field
k = GF(5)  # or use QQ for rationals

# For P^1, we work with the two standard affine patches
# Patch U0: where x0 != 0, coordinate t = x1/x0
# Patch U1: where x1 != 0, coordinate s = x0/x1

# Define the affine patches as polynomial rings
A0.<t> = PolynomialRing(k)  # O(U0) = k[t]
A1.<s> = PolynomialRing(k)  # O(U1) = k[s]

# The global sections on P^1 are just constants
O_X = k  # This is already a field

# Take a non-zero constant f = 2 (invertible since char != 2)
f = O_X(2)

# Localize at f: since f is already invertible, localization does nothing
O_X_f = O_x.fraction_field()  # Still k
print(f"Global sections localized at f: {O_X_f}")

# Compute X_f: since f is invertible everywhere, X_f = P^1
# The ring of functions on X_f is still just k
print(f"Functions on X_f: {O_X}")

# Verify the isomorphism
print(f"Are they equal? {O_X_f == O_X}")  # True
```

A more interesting case — affine line with doubled origin:

```
# Define the ring for the affine line
A.<t> = PolynomialRing(k)

# Localization at t (removing the origin)
A_local = A.localization(t)
print(f"Localized ring: {A_local}")
# Output: Fraction Field of Univariate Polynomial Ring in t over Finite Field of size 5

# This is k[t, t^{-1}]
print(f"Element example: {A_local(t^-2)}")  # 1/t^2
print(f"Another element: {A_local((1 + t)/(t^3))}")  # (1+t)/t^3

# Verify: O_X(X)_f = k[t]_t = k[t, t^{-1}]
# And O_X(X_f) = k[t, t^{-1}] (functions on A^1 \ {0})
# The proposition says these are isomorphic
```

Output:

```
Localized ring: Fraction Field of Univariate Polynomial Ring in t over Finite Field of size 5
Element example: 1/t^2
Another element: (t + 1)/t^3
```

What Sage is showing us

1. **For $\mathbb{P}^1$**: Since global functions are just the field k, localizing at any non-zero constant doesn't change anything. The proposition holds trivially.
2. **For the doubled affine line**: Sage computes $k[t]_t$ as the fraction field $k(t)$, which equals $k[t, t^{-1}]$ — exactly the ring of functions on $\mathbb{A}^1 \setminus \{0\}$. This verifies the proposition!

3. **Key insight**: Sage's `localization` method implements exactly the algebraic operation described in Proposition 3.12, and we can check that it matches the geometric description of X_f.

Limitation: Sage doesn't have built-in support for non-separated schemes like the affine line with doubled origin as a single scheme object — you have to work with the affine patches separately and glue them by hand. This is exactly why Proposition 3.12 is useful: it lets us work algebraically with $\mathcal{O}_X(X)_f$ instead of directly with the glued scheme X_f.